Frayer models

SECTION A / VOCABULARY Name _____

Find the word on the screen. Talk it through with your partner, then write or sketch in its four boxes.

Reference point

Meaning in my words How would you explain reference point to a friend?	**How can you tell?** A sign says “5 m east.” What place do we need to identify?
Example or sketch How would you mark a cart 3 m west of a doorway chosen as zero?	**Non-example and why** If I choose the table as zero instead, did the cart actually move?

Known quantity

Meaning in my words What makes a number a known quantity in a problem?	**How can you tell?** A cart starts at rest and is timed for 3 s. Which two values can you write down?
Example or sketch A sensor reads +4 m at the start. How would you enter that known value?	**Non-example and why** A friend lists a guessed final speed among the given values. Where should it go?

Frayer models

SECTION A / VOCABULARY Name _____

Find the word on the screen. Talk it through with your partner, then write or sketch in its four boxes.

Model assumption

Meaning in my words How would you explain model assumption to a friend?	**How can you tell?** What must be true to predict another 2 m each second from a cart's earlier motion?
Example or sketch Write an assumption beside a straight-line forecast for XJ.	**Non-example and why** Is tomorrow's camera reading an assumption once we measure it?

Range

Meaning in my words How would you explain range to a friend?	**How can you tell?** What positions are included in 12 ± 2 m?
Example or sketch How would you mark the forecast on a number line?	**Non-example and why** A partner says only 12 m counts as a match. What do the endpoints show?

Doodle the idea

SECTION B / DRAW WITH YOUR TEACHER Name _____

Sketch the reference frame and one sideways component. Box your time unit and one model assumption.



My marks mean

On assignment: Reference point

SECTION E / BACK / FINAL MEETING ONLY Name _____

Two observatories file the same sighting: “5 km right of station A” and “2 km left of station B.” B is 7 km right of A.

PICK A JOB / DELIVER ONE PRODUCT

☐ **Survey engineer**

Put both stations on one axis. Give dispatch one position in A’s frame, then list what else a future prediction needs.

☐ **Map broadcaster**

Make an on-air locator map. Convince the producer whether the two reports disagree by showing both zeros.

One partner makes the graphic; one delivers the first call. Show your evidence to the crew or the audience.

OUR GRAPHIC / COMMAND OR ON-AIR CUE

INCOMING UPDATE / WAIT FOR YOUR TEACHER

Listen to the new report. Switch speakers. Keep or change your call; point to the new evidence.